



BRIDGELABZ JAVA FULL STACK WITH REACT MAKER PLAN

Stages - Core Programming, Object-oriented Programming, DSA, Advanced Programming, Design, IO, DBMS, Backend Development, MicroServices, Web Dev, Frontend and Project Simulation

Fellowship Part	Num of Days	Fellowship Stage	Concept Introduced	App Developed
Core Programming	5	Core Programming Basics	Basic Elements of Programming: Variables, Literals, Primitive Data Types, Basic I/O, Operators Logical Constructs- condition (if.. else, ternary operator, Switch) and iterations (for, while, do while, for each), break, continue Debugging, Source Code Control : Git and GitHub Proper Programming Hygiene	- Java Environment Set up. All practice problems developed at this stage will be without any IDE - Basic I/O with different datatypes, Type conversions - Develop the OOPSBannerApp and Hello App applications. - Additionally, develop at least one of the following: 1. Guessing App or 2. SnakeLadder App.
		Core Programming Intermediate	Understanding JVM Architecture, Compiler and Interpreter, Garbage Collector Arrays (Single as well as Multi Dimensional), Copying Arrays (ArrayCopy()) Strings, Functions/ Methods, Pass by Value and Pass by Reference, Exception Handling	Develop a Gambling or Tic Tac Toe App using all core Java building blocks to demonstrate logical programming and functional design skills.
Core Programming	1	Workshop	QnA and Workshop on a problem given to ensure the skills learned are being applied.	
Object Oriented Programming	4	OOP- Basics	OOP Basics – Classes, Variables and Methods, Constructors, Access Modifiers Classes Vs Objects Differences, Instance vs Class Variables and Methods this and super keywords, InstanceOf and Static Operators, Eclipse IDE	Employee Payroll app with staff register, authenticate, and view their pay details (salary, deductions, taxes, net pay) securely.
		OOP-Advanced	OOPS Concept and Programming, Inheritance The Object class, Method Overriding or Overloading, Classes vs Interface Interface, Class vs Interface Abstraction, Polymorphism and Encapsulation	My Contacts App with users register, log in, and manage contacts (create, view, search, edit, delete). Supports tagging, notes, and optional sync/backup/export.
Object Oriented Development	1	Workshop	QnA and Workshop on a problem given to ensure the skills learned are being applied.	



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Data Structure & Algorithms	5	Data Structures	Need for Data Structures, Linked List (Single, Double, Circular) Stacks and Queues (Simple, Circular Priority) using Array and LinkedList Hash Map & Functions Tree and Graph -Overview Practical applications of different Data Structures	PalindromeChecker App: Uses Stack/Queue to push/enqueue string characters, compares stack.pop() with queue.remove() until empty (palindrome if matches); preprocesses for modes (case-insensitive/ignore spaces via toLowerCase/filter), handles batch inputs with explanations. BookMyStay App: Manages hotel inventory with HashMap (room types→counts/prices/amenities), uses PriorityQueue for first-come-first-served booking order, atomic updates via synchronized blocks prevent double-booking; handles reservations, add-ons (breakfast/spa), availability holds with explanations. Train App: Uses ArrayList/LinkedList for train consist (bogies sequence), HashMap for bogie details (type→capacity/cargo/safety), Enum for classes (Sleeper/AC/First) & shapes (Rectangular/Cylindrical); tracks composition, enforces constraints like petroleum-only cylindrical bogies.
		Algorithms	Sorting - Bubble, Insertion, Merge, Quick Searching - Linear, Binary Run Time Analysis, Big O notations Time and Space complexity of algorithms	
DSA	1	Workshop	QnA and Workshop on a problem given to ensure the skills learned are being applied.	
Advance Java Programming	8	Java Collections	Generics - Introduction to Generics, Type Parameters, Generic Classes, Generic Methods, Bounded Type Parameters, Wildcards Collection Framework - ArrayList, Linked List, Stack, Queue, Set, Map	Develop an BookMyStay App further : 1. Uses generics with type parameters, bounded types, wildcards for type-safe room handling; ArrayList for bookings, LinkedList waitlist, Stack undo reservations, Queue first-come-first-served, Set unique guests, Map roomType-to-inventory with atomic updates. 2. Handles checked exceptions for I/O operations, unchecked for invalid inputs, custom exceptions like NoRoomAvailableException extending Exception class with constructors; uses try-catch-finally for booking validation and stream cleanup, throw for immediate errors, throws in method signatures. 3. Java 8: Lambda expressions for comparators, Streams for filtering/sorting availability, Optional for null-safe guest lookups, method references in validation 4. Validates guest emails, password strength, phone formats using Pattern/Matcher with matches()/replaceAll(); JUnit tests booking scenarios with @Test annotations, assertions for availability/inventory; SonarQube ensures 85% + code coverage on validation/booking logic. 5. Uses @Override/@Deprecated for booking methods, @SuppressWarnings for legacy code, custom @BookingValidation annotation with elements for required fields; processes annotations at runtime for validation rules and method
		Exception Handling	Exceptions - Types of Exceptions: Checked Exceptions, Unchecked Exceptions and Custom Exceptions Handling Exceptions - Try-Catch Blocks, Finally Block, Throw vs. Throws, Creating Custom Exceptions, Extending Exception Class, Defining Constructors, Try-with-resource	
		Java 8 Features	Java 8: Streams, Lambda Function, Functional Interfaces, and Collection API updates.	
		Regex & JUnit	Regex - Common Patterns: Email Validation, Password Strength, Phone Number Format etc. Java Regex API: Using Pattern and Matcher, Methods: find(), matches(), replaceAll() Unit Testing with JUnit - Introduction to JUnit Framework, JUnit Annotations, Writing Test Cases, Structure of a Test Class, Using Assertions Code Coverage - SonarQube	



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		Annotaion	Annotation - Introduction to Annotations, Common Built-in Annotations, Creating Custom Annotations, Defining Annotations	interception.
Advanced Programming	1	Workshop	QnA and Workshop on a problem given to ensure the skills learned are being applied.	
Design Modeling	5	Design Principles	Design Principles like DRY, SOLID, YAGNI etc, Applying Design Principles and TDD Concept	Develop an Quantity Measurement App applying software design principles such as DRY, SOLID, and YAGNI.
		Design Patterns & Multi Threading	Intro to GoF Design Patterns, Different Kinds of Design Patterns Creational - Singleton, Factory, Builder Structural - Decorator, Behaviour - Observer Multi-Threading Concepts - Using Thread Class and Using Runnable Interface. Life Cycle and States of Thread in Java	Develop an Trading & Portfolio Tracking a fintech app that allows users to manage multiple investment portfolios (Stocks, Crypto, Mutual Funds). It calculates returns based on different strategies, provides real-time price alerts, and processes high volumes of trade orders using concurrent programming. Multithreading: Handling concurrency to prevent data race conditions, Executing asynchronous task
IO Program & DB	4	IO Programmimg	CSV Data Handling - Introduction to CSV, Reading CSV Files, Writing CSV Files, Parsing CSV Data, Data Validation JSON Handling - Introduction to JSON, Reading JSON Data, Writing JSON Data, Parsing JSON Data, JSON Schema Validation I/O Stream - File, Object, ByteArray, Buffered, Reader/Writer	IPL Analyser / Census Data Analyser App
IO Program & DB	4	Data Base	Different type of DBMS - Relational / Non-Relational. When to use what? Database: MySQL, RDBMS, DDL, DML, ER Diagram, Table Indexing, Normalization forms, Joins, Stored Procedures, Triggers. ER Diagram, JDBC	Health Clinic App: a console-based system designed for a small clinic to: - Register patients - Manage doctors and specialties - Schedule appointments - Track visit history and basic billing - Persist data using JDBC (MySQL)
Design, I/O & Multithreading	2	Workshop	IO Programming, ER Diagram, JDBC	



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Backend Development	5	Backend Basics	Tomcat and Servlet, Spring Intro and Spring Core Concept, Spring MVC, Spring REST API Calls and Request Handlings, H2 DB, Distributed Architectures, REST API Programming, REST Assure and JSON Server, exposure to SDLC	My Greetings App and Contacts app, Develop the backend of the Contacts app to learn the Concepts introduced daily in the Live Class.
		Backend w/ Spring Boot	Spring Boot, Spring Controller and REST API Dependency Injections, H2 Database Spring REST API, Spring Services, Spring JPA and Spring JDBC, Spring Scopes, Logging and Maven Postman	Employee Payroll App My Greetings App, Address Book App
Backend	1	Workshop	QnA and Workshop on a problem given to ensure the skills learned are being applied.	
Backend Development	4	Advance Backend Development	Spring Boot, Spring Security - JWT, JPA, JMS, Redis RabbitMQ, Spring Batch Excel	Fundoo Notes App : Backend token-based system developed for the fundooNotes application includes comprehensive modules: 1. User Management Module: Handling user login, registration, and password recovery. 2. Authentication & Authorization Module: Secure token-based verification for user access. 3. Notes Management Module: Functions to create and delete notes. 4. Pin / Archive / Trash Module: Allowing users to pin, archive, or delete notes for better organization. 5. Search & Filter Module: Enabling users to search and filter notes effectively. 6. Tags / Labels Management Module: Managing categorize notes with tags and labels. 7. Reminder & Notification Module: Setting and managing reminders with corresponding notifications. 8. File Attachment Module (optional): Supporting attachments to notes. 9. Asynchronous Operations via JMS: Ensuring smooth non-blocking background processes. 10. Token Caching via Redis: Enhancing authorization performance through token caching.
Backend Development		Advance Backend Development	Applying Spring REST API, Spring Scopes, Mapping Methods, Spring Boot Global Exception, Logging, AOP Spring Cloud	



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Backend Development	3	Microservices	Architecture : Monolith vs Microservices Spring Boot Microservices, API Gateway, Service Registry Eureka	Development of Microservices for fundooNotes Application: 1. User Management Microservice 2. Authentication & Authorization Microservice 3. Notes Management Microservice 4. Search & Filter Microservice 5. Tags / Labels Management Microservice 6. Reminder & Notification Microservice
Backend Development	1	Workshop	QnA andWorkshop on a problem given to ensure the skills learned are being applied.	
Web Programming	7	Web Programming Foundation	HTML5 , CSS3, DOM, Forms, Flexbox, Media queries	Develop an MyContacts Web App with users register, log in, and manage contacts (create, view, search, edit, delete). Supports tagging, notes, and optional sync/backup/export.
		Java Script	JS Programming :REPL, Data Types, Basic Programming Constructs, Array and Map Data Structures. Array functions like forEach, filter, map, reduce, Arrow Functions, Objects, Classes, Exception handling, Patterns	
Web Programming	3	Typescript Programming	TypeScript - OO Programming using TS, Types, Classes, Interface, Access Modifiers, Decorators, Exception Handlings, Logging, File IO, Source Control Management with Coding Standards	
Web Programming	1	Workshop	QnA andWorkshop on a problem given to ensure the skills learned are being applied.	
Frontend Development	5	Front End using ReactJS	ReactJS, JS, Virtual DOM, JSX, Form, Bootstrap, SCSS, Components, State and Props, Services, States and Lifecycle, Forms (Reactive & Template), Decorators, Events. Rendering, Axios, Events, Media Queries, Responsive UI, Authentication, Session management	fundooNotes Application: Part I: Develop frontend for fundooNotes App with ability to create users and allow them to login into system. Ability to create Notes and show it in the dashboard. User can perform CRUD operations on an existing notes with features of reminders, collaborators to be added over there. User can move a note into trash and can recover the same also.



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Frontend Development	5	Front End using ReactJS	Conditional Rendering, Lists and Keys, Hooks, Refs, Portal, Interceptors, Profiler HOC, Context, Render Props, Animations, Media Content, Environment Debugging using developer tools - Lighthouse	fundooNotes Application: Part II: Ability to create labels and checklist for notes. Complete App should be develop using React and integrate the REST APIs that have been developed in the BackEnd Development Stage.
Front end Development	1	Workshop	Day Long Workshop on a problem given to ensure the skills learned are being applied.	
Project Simulations	7	Book Store App	It begins with setting up the GIT Environment and then Full stack development of Bookstore Web based Application. This involves understanding SRS, breaking the problems into User Stories, developing and final project showcasing.	